

Latest in verification software: A bird's-eye view

MATLAB. Big Data enhancements (allowing Big Data analysis on desktop), new hardware support and integrated documentation for custom toolboxes

Simulink. Dashboard block library, Bus Smart Editing Cue, algebraic loop highlighting, Simulink project sharing, fast restart and simulation stepper for accelerated models, consistent data support for testing components and Simulink package for Apple iOS devices are some improvements

LabVIEW. Implementation of a local human machine interface (HMI) with embedded UI support, Ethernet over USB connection for simplified target configuration, debugging and maintenance, real-time trace viewer, USB 3.0 support and 12-core CPU support on Phar Lap ETS targets

Innovus. First massively-parallel implementation solution in the industry that supports advanced 16nm/14nm/10nm FinFET and established process nodes

Latest in PCB designing tools: A bird's-eye view

OrCAD. Static phase differential pair constraints, minimum/maximum and relative propagation, heads-up displays and delay tuning. Productivity improvements include scribble route, group/contour routing and via arrays

PADES. Available in the cloud or on the computer, PADS Standard Plus can be used for basic PCB layout needs and PADS Professional, which utilises Xpedition technology, can be used for schematic design, layout, constraint management, analysis, FPGA-PCB co-design, project data management and manufacturing prep in the most demanding and complex designs

CADSTAR. Contains across-the-board performance enhancements and design-efficiency features. CADSTAR's Design Editor now allows items to be modified within a group. PR.Editor Move tool in Activ-45 has been enhanced to use the pusher and spring-back engine

CR-8000. Engineering Desktop design portal enhancements, single product handling, hierarchical design and block reuse are some of the additions in the latest version

the existing plan, simulation vendor specific plans and annotated simulation results on to the plan."

There are some features that were earlier optional for users to use, but in the latest version these work as default. Singh informs, "Tool Command Language Application Programmer's Interface (TCL API) was optional in an earlier version, but it is now included by default. It helps users create their own outputs from the specification." He adds, "The latest version completely supports several industry-standard buses like Advanced eXtensible Interface (AXI), Advanced High-performance Bus (AHB), AH-B3Lite, Open Core Protocol (OCP), Avalon and Wishbone. This ensures that users can create SoCs with ease."

ADS. The newest release of Advanced Design System (ADS) 2015.01 includes various improvements over the earlier version. This tool from Keysight Technologies typically covers the complete design flow of a

product to manufacturing and is suitable for board, module and system designers.

Silicon radio frequency integrated circuit (RFIC) interoperability with Virtuoso enhancements provides designers with a schematic interoperable process design kit (PDK) as well as pcell support.

Improvements in finite element (FEM) simulation enable up to 16 times faster performance. There are also several layouts and layout verification improvements implemented in this version such as polymorphic interconnect modelling, layout interconnect design and faster layout viewing, to name a few.

Tools for system-level verification

Developers and integrators require verification tools to ensure that product specifications have been accurately tested and implemented. Using the latest in methodologies, new

developments in verification tools look to increase the overall quality of the final product.

MATLAB and Simulink. Engineers today are using FPGAs for prototyping and algorithm acceleration. These can be realised with the help of system-level design and verification tools like MATLAB and Simulink, by quickly prototyping their algorithms on FPGAs.

In the latest release of MATLAB, many new features have been included. It comprises a new graphics system and capability to analyse Big Data on a personal computer, after which it can be scaled to Hadoop.

Engineers can also pack custom MATLAB toolboxes into a single, installable file, and manage and control code within the MATLAB desktop.

Simulink's recent update enables engineers to test simulations with intuitive graphical displays and controls. With Bus Smart Editing Cue, it is now possible to combine signals into a bus signal and select specific signals or the entire bus can be accessed.

Project sharing and collaboration has been made easier with a share option. Users can share projects via GitHub, MATLAB toolbox and email, which is currently limited to Windows operating system (OS) only. Simulink also extends support to Apple iOS devices. This allows developers to create an app, which runs Simulink algorithms and models on iOS platforms.

LabVIEW. Engineers use system design tools to simplify complexity and for efficient hardware integration. Selecting the right design tool is the key to reduce development complexity and costs. Another popular tool, LabVIEW, recently got an upgrade that enables engineers to build better embedded monitoring and control systems, and make insightful data-driven decisions.

LabVIEW 2014 has a variety of enhancements made. For instance, in block diagrams, it automatically creates a shift register when a user